

GoodBelly case study

How to do the whole analysis in JASP

Every click, the logic behind it, the numbers you should see, and how to turn the output into the report

Tested on JASP 0.98 (macOS). Menu names can differ slightly in other versions.

Part	What it covers
A	Setup: the data file and the variables
B	The analysis, step by step (11 steps, each with logic and expected numbers)
C	What to export, and which output goes into which report section
D	How to write the report (outline, sentences, mistakes to avoid)
E	If something does not match

How to use it: do the steps in order. After each step, compare your numbers with the "Check against these numbers" line. If they match, you are on track; if not, jump to Part E.

Part A. Setup

A1. The data file

Start from **GoodBelly_JASP.csv** (in this folder). It is the original dataset plus a few extra columns:

Column(s)	What it is	Why it is there
Date, Store, Region, Week	identifiers	Week = 1 to 11
Units	weekly units sold (Y)	dependent variable
Price, Rep, Endcap, Demo, D13, D45, Natural, Fitness	the original variables (D13 = Demo1-3, D45 = Demo4-5)	the 8 predictors
Rep_x_Endcap, Rep_x_Demo, ... Fitness_x_Demo (16 columns)	products of two variables (interaction terms)	candidates for stepwise (Step 4)
D13_plus_D45	D13 + D45	needed for the "no fade" F-test (Step 7)
lnUnits	natural log of Units	only if you want to test a log model

If you would rather start from the original Excel dataset: add the interaction columns in Excel first (for example =Rep*Endcap), then save as CSV. The same approach gives D13_plus_D45 (=D13+D45).

A2. Load it

- Open JASP → open the CSV file (File → Open, or drag the file onto JASP).
- A "Data Preview" box appears. Leave the delimiter as comma and click **Load**.
- Check the data table: 1,386 rows. Units and Price should show a ruler icon (scale); Rep, Endcap, Demo, D13 and D45 show nominal icons; Natural and Fitness may show an ordinal icon. All of these work as regression predictors. If you want to be safe, click the icon at the top of Natural and Fitness and set them to Scale.

Concept: dummy variables are 0/1 columns, "no recent demo" is the base (S5 p.50-51, S6 p.11-13).

A3. Open the regression tool

Top toolbar → **Regression** → under **Classical** choose **Linear Regression**. The left panel holds the variable lists and option sections: Model, Statistics, Method Specification, Plots, Export. The right panel shows the output tables.

Part B. The analysis, step by step

The story of the analysis: fit an obvious model, check whether it is valid, find out why it fails, fix it, confirm the fix, then answer the case questions.

Step 1. Look at the data (optional but useful)

Click: Descriptives → Descriptives. Move Units, Price, Natural, Fitness into Variables. Tick Mean, Std. deviation, Minimum, Maximum. Under Plots tick Distribution plots.

Logic. Before modelling, know the shape and scale of Y and look for problems. Units are right-skewed (a few very high sales).

To see the endcap story early: put Units in Variables and Endcap and Rep in the **Split** box. You get the average units for each combination.

Check against these numbers. Units mean 253.8, SD 111, range 48 to 1,041. Group means: no endcap/no rep 198, no endcap/rep 276, endcap/no rep 218, endcap/rep 757.

Concept: raw averages hint at the story but cannot separate overlapping levers (S4 p.2-3, "control").

Step 2. Model 1: regress Units on all 8 variables

Click: Regression → Classical → Linear Regression.

- Move **Units** to "Dependent Variable".
- Select Price, Rep, Endcap, Demo, D13, D45, Natural, Fitness (click the first, hold Cmd and click the rest) and move them to "Covariates".
- Leave "Method" = **Enter** (all variables go in together).
- Open **Statistics** and tick: R squared change, F change, Durbin-Watson, Estimates, Confidence intervals (95%), Tolerance and VIF, Model fit, Part and partial correlations. Under Residuals tick Statistics and Casewise diagnostics (Std. residual > 3).
- Open **Plots** and tick: Residuals vs. predicted, Residuals histogram (with Standardized residuals ticked), Q-Q plot standardized residuals.

Logic. Model 1 is the natural first model and the baseline we will test. Every statistic ticked above feeds a later step: Durbin-Watson, residual plots and Q-Q for the assumption checks; Tolerance/VIF for multicollinearity; Part and partial correlations for understanding stepwise.

Check against these numbers. R^2 0.673, adjusted R^2 0.671, RMSE 63.7, $F = 353.7$, $p < .001$.

Coefficients: Price -28.54, Rep 77.44, Endcap 305.1, Demo 111.1, D13 73.52, D45 67.57, Natural -1.59 ($p = 0.37$), Fitness -1.02 ($p = 0.35$). Standardised column: Endcap 0.527, Rep 0.347, D13 0.241, Demo 0.235, D45 0.161, Price -0.119.

Concepts: multiple regression and output (S4 p.2-3, S2 p.26-29), overall F-test (S5 p.21), t-test for each coefficient (S3 p.17-23, S5 p.22), standardised coefficients (S5 p.30-34).

How to read it. The ANOVA F-test says the model is useful overall. Each coefficient has a t-test (H_0 : coefficient = 0); $p < 0.05$ means the variable matters. "Holding the others fixed, an endcap adds about 305 units a week." Natural and Fitness are not significant.

Step 3. Check Model 1's assumptions

Where: the same analysis. Read the Model Summary (Durbin-Watson column), the Coefficients table (VIF column), the residual plots and the Casewise diagnostics table.

Logic. The p-values from Step 2 are only trustworthy if the regression assumptions hold: errors with mean zero and constant variance, independent of each other, normally distributed, and no multicollinearity. A high R^2 does not guarantee that.

Assumption	What to look at in JASP	What a good model looks like
Linearity, constant variance	Residuals vs. predicted plot	points scattered evenly around zero, no funnel, no pattern
Independence	Durbin-Watson in Model Summary	close to 2 (below 2 = positive autocorrelation)
Normality	Histogram and Q-Q plot of standardised residuals; Residuals Statistics	Q-Q points on the diagonal; skew near 0, kurtosis near 3
No multicollinearity	VIF column in Coefficients	below 4 (tolerance above 0.25)
Outliers	Casewise diagnostics (std. residual > 3)	few cases; none with Cook's distance above 1

Check against these numbers. Durbin-Watson about 1.39; VIFs all between 1.02 and 1.26; the residual plot is uneven with a block of large negative residuals (down to about -370); the Q-Q plot has a heavy left tail (kurtosis about 8.5); about 23 cases beyond ± 3 . **Conclusion: Model 1 fails.**

Concepts: assumptions S3 p.45 and S5 p.20; residual plots S2 p.30, S3 p.34-43; Durbin-Watson S8 p.33-37; VIF S4 p.41-45.

Step 4. Find what is missing: stepwise regression with interactions

Click: add a new Linear Regression (Regression → Classical → Linear Regression again; the new analysis appears below the first).

- Move **Units** to "Dependent Variable".
- Set "Method" to **Stepwise**.
- In the variable list, select **Price** and then Shift-click **Fitness_x_Demo** to select all 24 candidates (the 8 original variables and the 16 interaction columns). Do not include Date, Store, Region, Week, D13_plus_D45 or lnUnits. Move them to "Covariates".
- Open **Method Specification**: "Use p value", Entry **0.05**, Removal **0.1**. These are the defaults, and they are the p-in and p-out of Session 7.
- Tick the same Statistics and Plots as in Step 2 (R squared change and F change matter most here).

Logic. Model 1 failed, so something is missing. An interaction (a product of two variables) lets one variable's effect depend on another. We suspect an endcap only works where a regional rep is present. Instead of choosing variables by eye, stepwise adds one variable at a time: at each step it picks the candidate with the largest partial correlation, adds it if its partial F-test gives $p < 0.05$ (p-in), then removes any variable whose p rises above 0.10 (p-out), and repeats until nothing changes (S7 p.31-33, p.54).

Step	Variable entered	R^2	R^2 change	F change
1	Rep_x_Endcap	0.548	0.548	1679
2	Rep	0.666	0.118	490.0
3	D13	0.722	0.056	277.2
4	Demo	0.769	0.047	278.3
5	D45	0.798	0.030	203.5
6	Price	0.806	0.007	52.56

Check against these numbers. JASP produces one model per step (M1 to M6). The R^2 values above must match. Read the note under the coefficient table to confirm that the final model contains Rep_x_Endcap, Rep, D13, Demo, D45 and Price. Plain Endcap, Natural, Fitness and the other 15 interactions never enter.

How to read the R² change. It is the new variation each variable explains. The Endcap × Rep interaction alone explains 55%. In Session 7 terms, R² change = (part correlation)², and also = partial r² × (1 – R² so far) (S7 p.42-47). Example: Rep has a partial r² of 0.262 and 0.452 was still unexplained, so it adds 0.262 × 0.452 = 0.118.

Concepts: interaction variables S6 p.48-54; part and partial correlation S5 p.12-19, S7 p.35-48; partial F-test S5 p.26-29, S7 p.51; forward and backward selection S7 p.57-58.

Why the interaction came first. Endcap × Rep is 1 only when a store-week has both an endcap and a regional rep (36 rows). The raw averages explain why it matters: endcap with a rep averages 757 units, endcap without a rep only 218 (almost the same as stores with no endcap, 198).

Step 5. Model 2: the final model

The easy way: the last step (M6) of the stepwise output is Model 2. Use its coefficient table.

To produce a clean Model 2 table: run another Linear Regression with Method **Enter**, Units as dependent, and covariates Rep_x_Endcap, Rep, Demo, D13, D45, Price. Tick the same Statistics and Plots as in Step 2.

Logic. Stepwise tells us which variables to keep; refitting them with Enter gives one clean table, with no intermediate steps, that goes into the report.

Term	Coefficient	Std. error	t	95% CI	Std. beta	VIF
(Intercept)	276.58	12.16	22.7	252.7 to 300.4	-	-
Rep_x_Endcap	454.42	8.47	53.7	437.8 to 471.0	0.651	1.05
Rep	59.44	2.99	19.9	53.6 to 65.3	0.267	1.27
Demo	106.76	5.70	18.7	95.6 to 117.9	0.226	1.03
D13	73.38	3.76	19.5	66.0 to 80.8	0.240	1.08
D45	74.56	5.03	14.8	64.7 to 84.4	0.178	1.02
Price	-22.06	3.04	-7.3	-28.0 to -16.1	-0.092	1.15

Check against these numbers. R² 0.806, adjusted R² 0.805, RMSE 49.0, F = 954, all p < .001. **Model 2: Units = 276.6 + 454.4·(Endcap×Rep) + 59.4·Rep + 106.8·Demo + 73.4·D13 + 74.6·D45 – 22.1·Price.**

How to read it. Endcap × Rep (454): an endcap adds about 454 units a week, but only in stores that also have a regional rep. Rep (59): a rep on its own adds about 59 units. Demo (107): the demo week. D13 (73) and D45 (75): each week after a demo. Price (-22): each extra \$1 removes about 22 units. The intercept (277) is the prediction at price \$0, which is outside the data, so we do not interpret it (S8 p.9-11).

Step 6. Check Model 2's assumptions

Same places as Step 3, on the Model 2 analysis.

Check	Model 1	Model 2 (what you should see)
Residuals vs predicted	uneven, big negative block	centred on zero, even spread
Durbin-Watson	1.39	2.11
Skew / kurtosis	-0.93 / 8.5	-0.01 / 3.05
Largest VIF	1.26	1.27
Cases with std. residual > 3	23	5
Q-Q plot	heavy left tail	points on the diagonal

Logic. A model is only worth reporting if it passes the checks that the earlier one failed. Model 2 passes all of them. The earlier problems came from the missing interaction, not from the scale of sales, so no log transformation is needed.

Outliers in more depth. Under Statistics → Residuals, change "Casewise diagnostics" to tick **Cook's dist., Leverage, DFFITS**. A Cook's distance above 1 marks an influential observation (S8 p.24); the leverage cut-off is $2(k+1)/n = 0.010$ (S8 p.27). Expect a largest Cook's distance of about 0.02, so no observation is influential.

Concepts: assumptions S3 p.45; outliers, leverage, Cook's distance S8 p.16-31.

Step 7. Partial F-tests: does the data support the case questions?

A **partial F-test** compares a smaller model with a bigger one (S5 p.26-29). In JASP you build it with **model blocks** (nested models in one analysis).

- Open a new Linear Regression. Move Units to "Dependent Variable". Keep Method = **Enter**.
- Open the **Model** panel and add a second model with the "+" (add new model) button. Put the starting terms in Model 1 and the extra terms in Model 2. (I did not test this panel in detail, so if it looks different in your version, look for "Add new model" or "Block".)
- Under Statistics tick "R squared change" and "F change".
- In the Model Summary, read the second row (M2): **F Change** and **p** are the partial F-test for the terms you added.

Logic. $F = [(SSE \text{ reduced} - SSE \text{ full}) \div q] \div [SSE \text{ full} \div (n - k - 1)]$, where q is the number of added terms. A small p means the extra terms really help; a large p means the smaller model is enough.

Question	Model 1 terms	Add in Model 2	Expected F change	p
Is the interaction essential?	Price, Rep, Endcap, Demo, D13, D45, Natural, Fitness	Rep_x_Endcap	944	<0.001
Does the demo lift fade? (D13 = D45)	Rep_x_Endcap, Rep, Demo, **D13_plus_D45**, Price	D13	0.04	0.85
Is plain Endcap needed?	Rep_x_Endcap, Rep, Demo, D13, D45, Price	Endcap	about 0	0.96
Do Natural and Fitness matter?	same as above	Natural, Fitness	0.04	0.96
Does Region matter?	same as above	Region (add as a "Factor")	0.89	0.54

Why the "no fade" trick works. D13_plus_D45 forces both effects to be equal. Adding D13 on top lets them differ, so the F-test asks whether letting them differ helps. It does not ($p = 0.85$); the lift is the same in weeks 1-3 and 4-5+.

Concepts: partial F-test S5 p.26-29; interpreting dummy coefficients S7 p.29.

Step 8. Demo payoff, by hand

JASP gives the coefficients; the payoff is simple arithmetic you can do in Excel. A demo is followed by the demo week (1 week), then weeks 1-3 (3 weeks) and weeks 4-5 (2 weeks):

Extra units per demo = $106.8 + 3 \times 73.4 + 2 \times 74.6 = \mathbf{476 \text{ units over six weeks}}$. Only 22% falls on the demo day, so judging a demo by demo-day sales understates it about four times.

Step 9. What a regional rep and an endcap are worth together

An endcap in a rep store: $59.4 + 454.4 \approx \mathbf{514 \text{ units a week}}$ over a store with neither. An endcap in a non-rep store: plain Endcap is not in the model ($p = 0.96$ when added back), so about **0**.

Step 10. Price elasticity (optional)

Elasticity at the mean $\approx b \times \text{mean price} \div \text{mean units} = -22.1 \times 4.11 \div 253.8 \approx \mathbf{-0.36}$. A 10% price cut raises units only about 3.6%, so revenue falls. (A log-log model is the class method for elasticity, S2 p.47; the point calculation above is simpler.)

Step 11. Save everything

Save the JASP file (File → Save As → GoodBelly.jasp) so the analysis can be reopened. Export the results next (Part C).

Part C. What to export and where it goes in the report

Export: File → Export Results (HTML or PDF, whichever your JASP version offers). To save a single plot or table: right-click it and choose "Save image as" or "Copy". Take a screenshot if the right-click menu is unavailable.

JASP output	Use it for	Report section
Descriptives, group means (Step 1)	Data description, raw hints	1. Understanding the problem and data
Model 1 Model Summary, ANOVA, Coefficients (Step 2)	Baseline model table	2.1 Baseline model (Question 1)
Stepwise Model Summary (Step 4)	Path table: step, variable, R^2 , R^2 change	2.2 Stepwise (Question 1)
Model 2 Coefficients with CI and VIF (Step 5)	Final model table and equation	2.2 (Question 1)
Residuals vs predicted, histogram, Q-Q for Model 1 and Model 2	Residual diagnostic figures	3. Assumptions (Question 2)
Durbin-Watson, VIF, Casewise diagnostics	Assumption comparison table	3. Assumptions (Question 2)
Partial F-tests (Step 7)	Duration of demo lift, placement	4 and 5 (Questions 3 and 4)
Step 8-10 arithmetic	Payoff, elasticity	4 and 7 (Questions 3 and 6)

Part D. How to write the report

The assignment (Deliverables and Questions.pdf) requires: **(1)** a cover page with title and team names; **(2)** a one-page executive summary; **(3)** a main report of at most 15 one-sided pages with the problem, regression models, graphs (residual and P-P/Q-Q plots), results, conclusions and recommendations for each question; total at most 25 pages; **(4)** all files in a "Group #.zip" on Moodle before **6 October, 9:00 AM IST** (no extensions); **(5)** the peer evaluation form, filled individually.

D1. Outline

Section	Content	JASP source
Cover	Title, course, group number, five names and roll numbers	-
Executive summary (1 page)	Problem, models, key findings, recommendations	all
1. Problem and data	Case, variable table, data checks	Step 1
2. Question 1	Model 1, stepwise path, Model 2 table and equation, interpretation	Steps 2, 4, 5
3. Question 2	Assumption table and plots for Model 1 and Model 2; what changed	Steps 3, 6
4. Question 3	Demo effects, no-fade test, payoff	Steps 5, 7, 8
5. Question 4	Endcap $\times$ Rep group table and result; other factors tested	Steps 1, 5, 7, 9
6. Question 5	What else was tested (Natural+Fitness, Region, interactions); suggestions	Steps 4, 7
7. Question 6	Recommendations and limitations	all
Appendix	Model 1 coefficient table; the JASP file and exported output	-

D2. The numbers to put in the executive summary

- Model 1 (all 8 variables): R^2 0.67, but Durbin-Watson 1.39 and a heavy left tail, so it fails the assumptions.
- Stepwise with interaction terms selected Endcap $\times$ Rep first; Model 2 has R^2 0.81, Durbin-Watson 2.11, near-normal residuals.
- Endcap adds about 454 units a week only in stores with a regional rep; with no rep the effect is about zero.
- A demo adds about 107 units in the demo week and about 73-75 a week for at least 5 weeks afterwards (no fade, $p = 0.85$); about 476 extra units per demo over six weeks.
- A regional rep alone adds about 59 units; price is inelastic (about -0.36); Natural, Fitness and Region do not matter.
- Recommendations: pair endcaps with regional reps; keep demos and judge them over six weeks; do not use price cuts for volume; test endcaps in a few more non-rep stores.

D3. Sentences you can adapt

- "Model 1 explains 67% of the variation, but its residuals are uneven and left-skewed and the Durbin-Watson statistic is 1.39, so the assumptions of regression are not met."
- "We used forward stepwise regression (p -in 0.05, p -out 0.10) on the eight variables and sixteen interaction terms. The Endcap $\times$ Rep interaction entered first and alone explains 55% of the variation."

- "Holding the other variables fixed, an endcap adds about 454 units per store-week in stores with a regional sales rep. In stores without a rep, plain Endcap is not significant ($p = 0.96$)."
- "A partial F-test comparing the lift in weeks 1-3 with weeks 4-5 gives $p = 0.85$, so we cannot reject that the two are equal: there is no fade within five weeks. We only know the lift lasts at least five weeks, because the last category is open-ended."
- "These results are associations, not proven causes: demos, endcaps and reps were not randomly assigned, and the no-rep endcap result rests on three stores."

D4. Mistakes to avoid

- Do not report only R^2 . Show the assumption checks and say plainly that Model 1 fails them.
- Do not call the three demo dummies "mutually exclusive": 34 store-weeks have more than one flag set because stores held repeat demos.
- Do not interpret the intercept (price = \$0 is outside the data).
- Do not say the endcap works everywhere. It works with a regional rep.
- Do not claim causation; use the word "associated".
- Keep the main report within 15 pages; put extra tables in the appendix.
- Fill in the cover page names, name the zip with your group number, and use your own words (the course checks originality with Turnitin).

Part E. If something does not match

Problem	What to check
Stepwise picks different variables	Make sure only the 24 candidates are in Covariates (not Week, lnUnits or D13_plus_D45), Method = Stepwise, Entry 0.05, Removal 0.1
Model 1 numbers differ	The 8 covariates must be Price, Rep, Endcap, Demo, D13, D45, Natural, Fitness. Check the CSV has 1,386 rows
A dummy shows as "Factor" in the way	Move it to Covariates, not Factors (Region is the only Factor)
Durbin-Watson is different	The rows must stay sorted by Store then Week, as in the CSV. Re-sorting changes it
The Model panel looks different	Look for "Add new model" or "Block". Alternatively run the two models as separate analyses and compute F by hand: $F = [(SSE_1 - SSE_2) \div q] \div [SSE_2 \div (n-k-1)]$; SSE = Residual sum of squares in the ANOVA table
Plots are hard to read	Right-click a plot → Save image as; or increase the JASP window size

Reference values for the whole analysis (use these to verify every table): Model 1 R^2 0.673, F 353.7.

Stepwise R^2 path 0.548, 0.666, 0.722, 0.769, 0.798, 0.806. Model 2 R^2 0.806, adjusted 0.805, RMSE 49.0, F 954. Durbin-Watson 1.39 → 2.11. Largest VIF 1.27. Partial F: interaction 944; D13 = D45 0.04 (p 0.85); plain Endcap p 0.96; Natural + Fitness p 0.96; Region p 0.54.